| INCH-POUND |

MIL-P-63514A (AR)
AMENDMENT 8
23 December 1994
SUPERSEDING
AMENDMENT 7
15 August 1994

MILITARY SPECIFICATION

PROJECTILE, 105MM, TP-T, M489A1,
METAL PARTS ASSEMBLY

This amendment forms a part of Military Specification MIL-P-63514A (AR), dated 12 February, 1986, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1: Delete:

"MIL-STD-649 - Aluminum and Magnesium Products Preparation for Shipment and Storage".

PAGE 3

2.2: Add:

"ASTM B-660 - Packaging/Packing of Aluminum and Magnesium Products."

PAGE 4

3.6.2: Delete in its entirety and substitute:

"3.6.2 Target-impact-dispersion. The pooled probable error (PE) shall not be greater than 0.24 mils horizontal and 0.22 mils vertical when tested in accordance with paragraphs 4.5.3 to 4.5.3.5. See 6.6 for method of calculating pooled PE."

PAGE 6

4.3.2 Title: Delete "Abrasion resistance." and substitute "Abrasion resistance (all unsealed type III coatings)."

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PAGE 8

Table I, Body: After Abrasion test, add "(Note 5)."

PAGE 9

Add Note 5: "5. All unsealed type III coatings."

PAGE 16

Note 1: Delete "See 4.4.3.2.1 for sample size" and substitute

"See 4.4.3.2.1 for sample size. (All unsealed type III coatings.)"

PAGE 20

4.4.2.4:

Major 102: Delete in its entirety and substitute:

"102 Projectile outside diameter, max. (Note 3) 100% 3.1 Ring Gage".

Add: "Note 3. To be performed immediately prior to packaging."

Major 103: Delete in its entirety and substitute:

"103 Rear obturator band diameter max 3.1 100 % Ring Gage".

Major 107: Delete in its entirety.

Change Category numbers:

From "108" to "107".
From "109" to "108".
From "110" to "109".
From "111" to "110".
From "112" to "111".

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PAGE 24

4.4.3.2.1 Delete the paragraph in its entirety and substitute:

"4.4.3.2.1 Abrasion resistance test for all unsealed type III coatings (body). After the protective coating has been applied to the bodies, two bodies (a specimen taken from the rear end of each body shall be used) selected from each production lot shall be subjected to the abrasion test as specified in MIL-A-8625. Failure of one or more specimen (taken from the coated body) to comply with the applicable requirements of MIL-A-8625, shall be cause for rejection of the production lot."

4.4.3.3 Add the following to 4.4.3.3: "Production lot salt spray test samples shall comply with paragraph 6.12."

PAGE 25

4.4.3.4.1: Add "(Excess pressure test for new manufacturers or when there is a break in production.)"

4.4.3.4.3: Delete in its entirety and substitute:

"4.4.3.4.3 Target impact dispersion. Samples provided in accordance with Table III and Table IV shall be subjected to this test. If the pooled probable error (PE) of the samples, calculated in accordance with paragraph 6.6 using the normal standard deviation method multiplied by the appropriate factor of Table II exceeds the requirements (see 3.6.2), then the standard deviation (SD) shall be calculated using the successive difference method (see 6.6.4). Select the SD with the smaller value for each group from the two SD calculations and then calculate the pooled PE value. If the pooled PE multiplied by the appropriate factor of Table II meets the requirements, the disposition of the lot shall be referred to the technical agency. If the requirements are still not met, the lot shall be rejected."

PAGES 26 and 27

4.4.3.4.4: Delete TABLE III in its entirety and substitute included TABLE III:

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"TABLE III. First article ballistic test.

Phase	Degree F	Wear	Qty	Requirement	Accept	Refer	Reject
A	+70	New	25	a. Security	0	-	1
				b. Flight - erratic	0	-	1
				c. Target-Impact-Dispersion (Note 2)			
				d. Velocity, Pressure, Spin Rate, for info only			
B	+70	Worn	10	a. Security	0	-	1
				b. Flight - erratic (Note 1)	1	-	2
				- target miss	1	-	2
				c. TID, Velocity, Pressure, Spin Rate, for info only			
C (NOTE 3)	Excess pressure +70	New	10	a. Security	0	-	1
				b. Flight - erratic (Note 1)	1	-	2
				c. TID, Velocity, Pressure (4.4.3.4.1), Spin Rate, for info only			
D	+125	New	15	a. Security	0	-	1
				b. Flight - erratic (Note 1)	1	-	2
				- target miss	1	-	2
				c. TID, Velocity, Pressure, Spin Rate, for info only			

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Phase	Degree F	Wear	Qty	Requirement	Accept	Refer	Reject
E	+125	Worn	10	a. Security	0	-	1
				b. Flight - erratic (Note 1)	1	-	2
				- target miss	1	-	2
				c. TID, Velocity, Pressure, Spin Rate, for info only			
F	-35	New	15	a. Security	0	-	1
				b. Flight - erratic (Note 1)	1	-	2
				- target miss	1	-	2
				c. TID, Velocity, Pressure, Spin Rate, for info only			
G	-35	Worn	10	a. Security	0	-	1
				b. Flight - erratic (Note 1)	1	-	2
				- target miss	1	-	2
				c. TID, Velocity, Pressure, Spin Rate, for info only			

NOTES:
1. All ballistic phase results, added together with the exception of the 70°F new tube phase, shall reject the first article sample if: three or more erratic flight total occurred, also, a combination of five or more erratic flights and targets missed occurred.

2. Target-impact-dispersion shall be tested in accordance with 4.4.3.4.3.

3. Phase C is not needed for current manufacturers."

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PAGE 28

Table IV. Production lot ballistic test: Delete excess pressure +70°F phase for current manufacturers.

PAGE 29

4.5.2.1 Title: Delete "Abrasion resistance." and substitute "Abrasion resistance (all unsealed type III coatings)."

* 4.5.3.1 Delete in its entirety and substitute the following:

"4.5.3.1 Ballistic phases. The metal parts assemblies shall be subjected to ballistic testing at the phases indicated in Table III (First Article) and Table IV (Production). The metal parts assemblies shall be assembled into complete rounds utilizing the service charge and shall be fired from a fixed-type facility mount with an M1 or M60A3 recoil utilizing a muzzle boresight or from an M1 or M60A3 tank with a gunner's auxiliary sight. An M68A1 gun tube shall be utilized as specified in Table I. All fixed type facility mounts and tank systems must be properly maintained and validated prior to testing."

PAGE 34

5.1.2.1: Delete "MIL-STD-649 Level C" and substitute "ASTM B-660 Commercial Packaging".

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PAGE 37

6.6.2: Change title to "Method for calculating normal standard deviation."

Add new paragraph 6.6.4:

"6.6.4 (U) Method of calculating standard deviation by successive difference. In using successive difference, the data must be arranged in the order of firing or occurrence. Truncation is not allowed and all valid data points shall be utilized in the calculation. The following equation shall be used when successive difference is applicable.

$$SD = \sqrt{\frac{\sum_{i=1}^{n-1} (x_{i+1} - x_i)^2}{2(n-1)}}$$

where:

SD = Standard Deviation

n = Sample size

x_i = Individual data values in one direction (horizontal or vertical)

x_{i+1} = Individual data values immediately following x_i".

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PAGE 38

Add 6.12:

"6.12 Salt spray test specimens. Salt spray samples shall be components/assemblies randomly selected from each inspection lot. The contractor may elect to use reject components/assemblies for salt spray samples. If the contractor elects to use rejected components/assemblies, a request shall be forwarded to the contracting officer and approval received prior to use. A detailed plan describing the selection and processing of the sample shall accompany the request and shall serve as the basis for approval. The plan must clearly establish the relationship of the test specimens to the overall manufacturing process and lotting criteria which will apply to the components being accepted. Specifically, the plan shall address how test specimens are selected and identified; how they are processed at the same time and in the same way as the lot they represent; and how they are randomly introduced and distributed when the protective finish is applied. In addition, procedures shall be included for the contractor to require his quality control staff together with the resident DCAS quality assurance representative, to maintain surveillance over the process to ensure test specimens are not lost or given preferential treatment."

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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